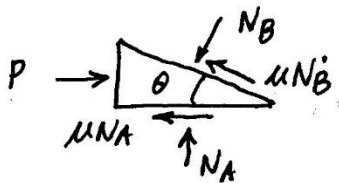
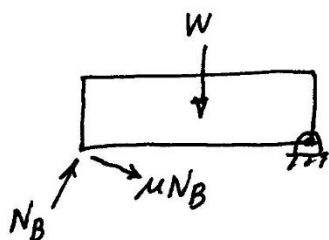


Ch 8 Friction: Wedges

- A wedge is a simple machine
- Often confers mechanical advantage



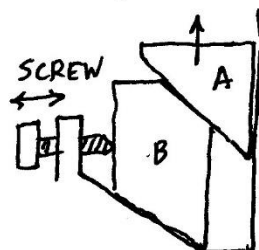
At small angles θ , a small value of P can exert a large force N_B to lift or level a heavy machine.



This ratio $\frac{N_B \text{ (Output)}}{P \text{ (Input)}}$ force is called Mechanical Advantage.

- Many wedges are "self-locking", meaning friction will keep them in place, keep them from "popping out."
- With a "self locking" wedge, can hit with hammer (large impulsive force) to lift heavy loads in small increments.
- A wedge converts $\rightarrow$ motion to $\uparrow$ motion.
- Large x change produces small y change.

• See HW:



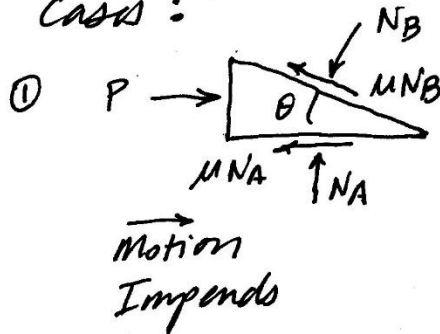
} Converts x motion to y motion.

- Usually treat wedges as particles: No $\sum M = 0$ equation, only write $\sum F_x = 0$
 $\sum F_y = 0$
- Usually assume wedges massless.

Wedges (cont'd)

- Important to study the problem.

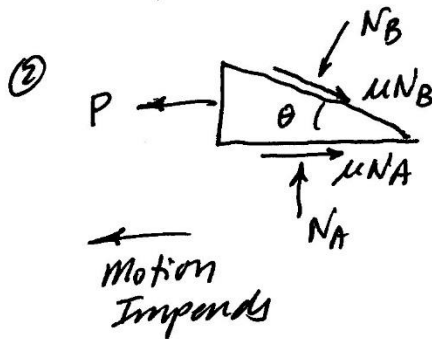
- Cases: (SINGLE WEDGES)



Lifting Case
P pushing wedge,

wedge used to lift or level a load.

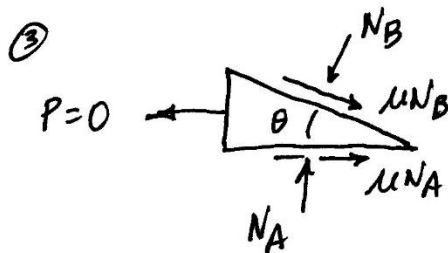
Frictions μN_A and μN_B act opposite of impending motion.



Lowering Case
P pulling wedge out,

used to lower a machine or load.

Frictions μN_B and μN_A act opposite



Lowering Case
P = 0

This is the "self locking" condition

All of these can be paired with another wedge.

Lifting Case

